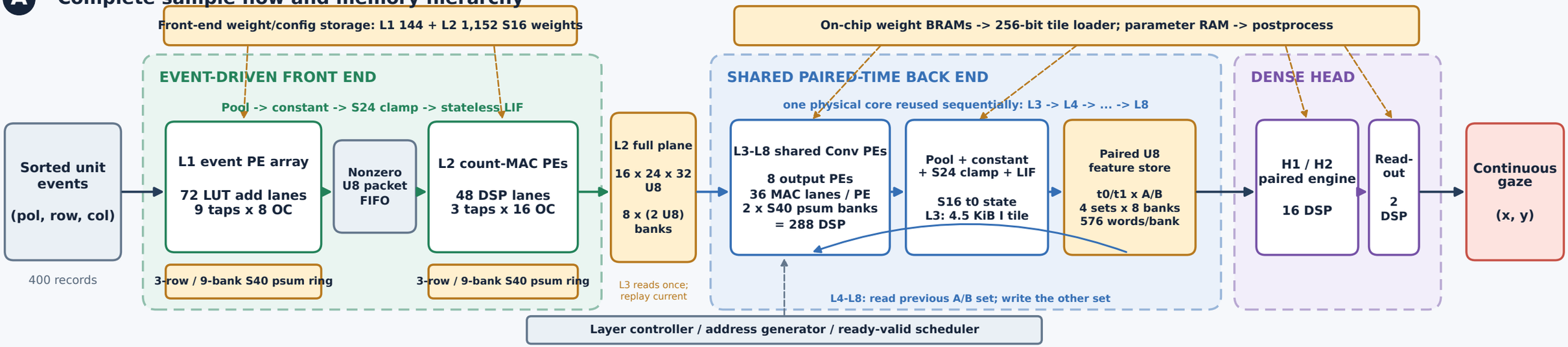
EBV-Eye Final RTL: PE and On-Chip-Memory Dataflow

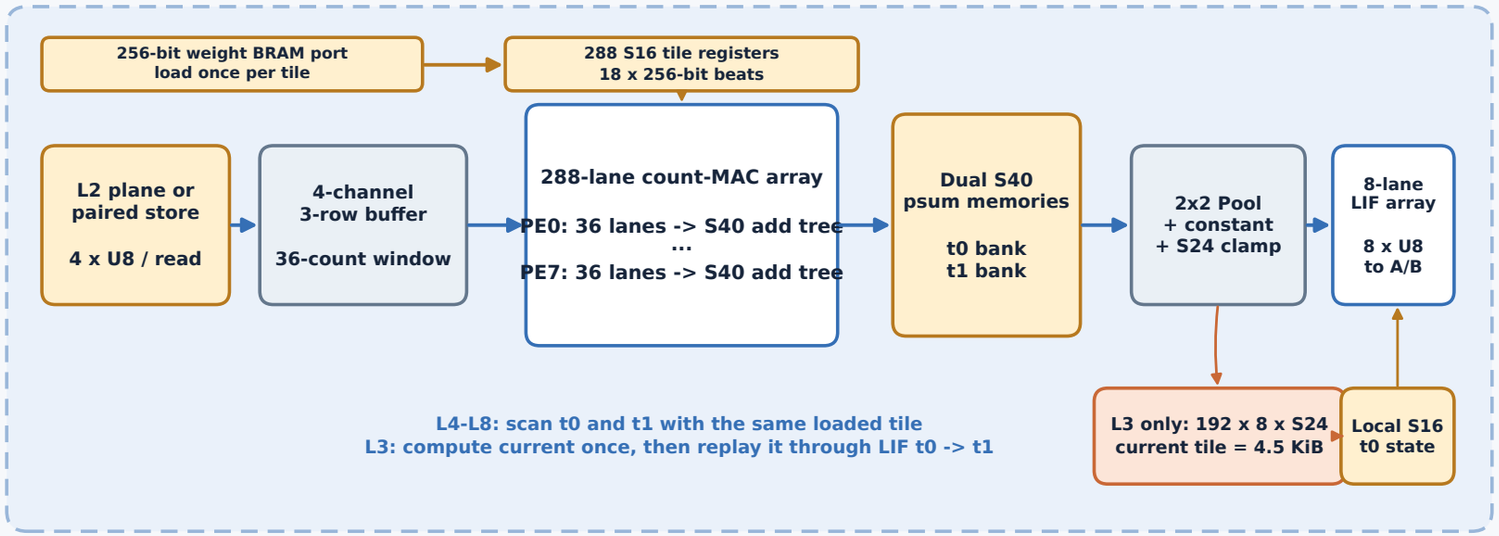
400 events | T = 2 | S16 W | S40 -> S24

Event-driven L1/L2, one shared 288-lane deep core, and adjacent t0/t1 execution

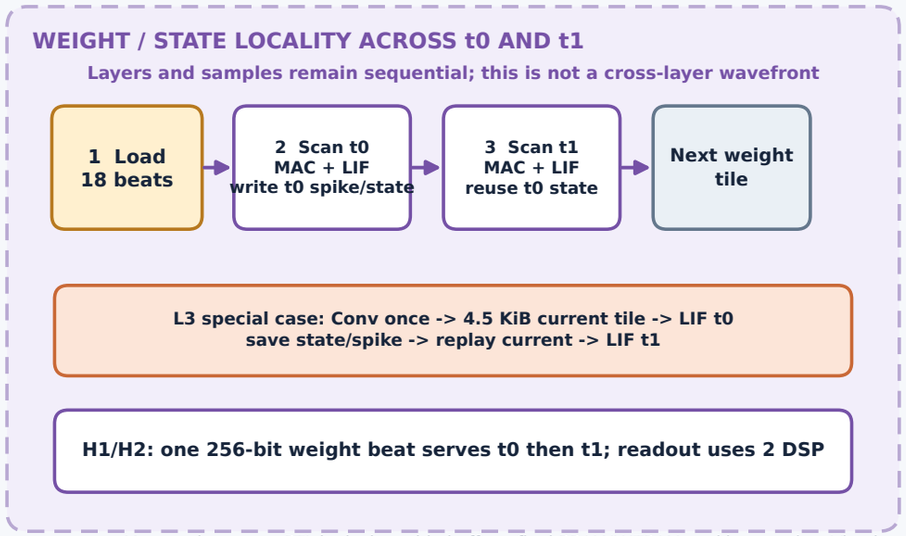
A Complete sample flow and memory hierarchy



B One deep-convolution tile: PE lanes and local buffers



C Per-tile execution order



Memory boxes are RTL logical on-chip buffers; final BRAM/LUTRAM packing requires Vivado.

Measured RTL/Yosys evidence | 354 DSP = 48 + 288 + 16 + 2 | 6,060 runtime 256-bit weight reads | 67,942.9 cycles/sample | 80.5 BRAM36-equivalent | 159,408 checked values, 0 mismatch